

# Agent Trust Fabric

## Cryptographic Authority Governance for Autonomous AI Agents



OMNIX QUANTUM

**Document:** OMNIX-WP-ATF-2026-001

**Version:** 1.0.0

**Date:** May 2026

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**Protocol:** RFC-ATF-1 · ADR-156/157/158

**Classification:** Public — Institutional Distribution

**Cryptography:** ML-DSA-65 (NIST FIPS 204 / CRYSTALS-Dilithium)

**Formal Methods:** TLA+ Model Checked — 5 invariants

**Repository:** [github.com/Costenho19/rfc-atf-1](https://github.com/Costenho19/rfc-atf-1)

### Abstract

Autonomous AI agents increasingly make consequential decisions in enterprise systems — executing trades, authorizing medical procedures, routing logistics, and managing critical infrastructure. Yet the governance of *who authorized these agents, under what authority, and whether that authority was valid at the moment of execution* remains unaddressed by existing frameworks.

This paper presents the OMNIX Agent Trust Fabric (ATF) — a formally specified, post-quantum cryptographic protocol that answers four questions for every AI agent execution: (1) Who authorized this agent? — via ML-DSA-65 signed Delegation Receipt; (2) What authority did it hold? — via Monotonic Authority Reduction, TLA+ verified; (3) Was the authority valid at execution? — via Temporal Admissibility Record captured before the pipeline runs; (4) Is the full chain independently verifiable? — via offline CLI verifier, no platform access required.

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# 1. Problem Statement

## 1.1 The AI Agent Authorization Gap

Modern AI governance frameworks address *what decisions are made* but not *who authorized the agent that made them*. When an enterprise AI agent executes a high-stakes action, the audit record typically captures:

- What decision was made
- What signals influenced the decision
- A cryptographic hash of the decision receipt

What is systematically missing:

- **Proof that the agent was authorized to act** — not just authenticated, but explicitly authorized with bounded scope
- **Proof that the authority was valid at the nanosecond of execution** — not valid yesterday, not valid in principle, but valid *then*
- **A verifiable chain from the AI agent back to a human principal** — traceable without trusting the platform

This gap is not a limitation of existing governance frameworks — it is not their design target. OAuth 2.0 handles access delegation; W3C Verifiable Credentials handle identity claims; JWT handles session validity. None was designed to answer: *"Was this autonomous AI agent's authority mathematically bounded and verified at the moment it executed this specific governance decision?"*

## 1.2 The Accountability Requirement

Regulation is closing this gap:

| Regulation         | Requirement                                                  | ATF Response                           |
|--------------------|--------------------------------------------------------------|----------------------------------------|
| EU AI Act Art. 13  | Transparency for high-risk AI — "who authorized this system" | ATF DR chain to Tier-1 human           |
| EU AI Act Art. 14  | Human oversight — human principal must be identifiable       | ATF chain_root_id → Tier-1             |
| DORA Art. 9        | ICT resilience — documented controls for automated systems   | ATF TAR proves control at execution    |
| MiCA Rec. 65       | Algorithmic trading controls — authorization audit trail     | ATF triple chain for trading decisions |
| NIST AI RMF GOVERN | AI decisions traceable to accountable entities               | ATF cryptographic traceability         |
| SOC 2 CC6          | Logical access controls — non-repudiation                    | ATF PQC signatures                     |

## 1.3 Why Existing Protocols Are Insufficient

| Protocol       | What It Provides          | What It Cannot Provide                                                                   |
|----------------|---------------------------|------------------------------------------------------------------------------------------|
| OAuth 2.0      | Access token delegation   | Monotonic authority bound; pre-execution temporal record; human chain traceability       |
| JWT (RFC 7519) | Signed claims with expiry | Authority budget arithmetic; separate admissibility artifact; offline chain verification |
| W3C VC         | Signed identity claims    | Authority bounds; temporal admissibility; mandatory cross-domain reduction               |
| SPIFFE/SPIRE   | Workload identity         | Authority delegation chain; governance receipt integration                               |
| OpenID Connect | Authentication            | Authorization chain governance; PQC signatures; formal invariants                        |

None of these protocols is wrong — they solve different problems. ATF addresses the gap that exists at the intersection of all of them: **authority governance for autonomous AI agents, before and during execution.**

## 2. The ATF Protocol Architecture

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### 2.1 Core Concepts

#### Agent Identity Record (AIR)

Every AI agent registered in an ATF-compliant system receives a globally unique identifier: `AID-{DOMAIN}-{16HEX}`. The AIR contains the agent's authority budget cap, domain, registration tier, and public key for verification.

#### Delegation Receipt (DR / ATFD<sub>R</sub>-{16HEX})

A PQC-signed record issued by a principal (human or agent) to an agent, explicitly granting a bounded subset of the principal's authority for a defined task scope. Content-hashed (SHA-256 over canonical JSON), signed with ML-DSA-65 (Dilithium-3, NIST FIPS 204), and independently verifiable offline using the embedded delegator public key.

#### Temporal Admissibility Record (TAR / ATFTAR-{16HEX})

A PQC-signed record issued at the exact moment of admission to the governance pipeline — **before any governance logic executes**. The TAR proves: (1) the DR was ACTIVE at `execution_ns` (nanosecond-resolution); (2) the admission decision was ADMITTED or REJECTED; (3) the `execution_ref` binds this TAR to a specific GovernanceReceipt.

#### Trust Lattice

The directed acyclic graph (DAG) of all delegation receipts. The lattice enforces the Acyclicity invariant (ATF-INV-002) and provides chain traversal for complete authority verification.

#### GovernanceReceipt

The governance decision record (ADR-028), extended with `atf_context` embedding the DR, TAR, and trust summary — forming the three-artifact audit chain.

### 2.2 The Authority Lifecycle

#### Phase 1 – Registration

Human Tier-1 registers agent → AIR created → authority budget assigned

#### Phase 2 – Delegation

Principal signs DR → `authority_budget_granted ≤ authority_budget_delegator`

DR stored in trust lattice → `chain_root_id` set

#### Phase 3 – Admission

Agent presents DR at execution boundary

`ATFConnector.admit()` called BEFORE governance pipeline

```

TAR issued: execution_ns captured, DR status verified, TAR signed
TAR.execution_ref = GovernanceReceipt.receipt_id

Phase 4 – Execution
Governance pipeline runs (AVM, AI, veto chain)
Decision produced with PQC-signed GovernanceReceipt
ATF context embedded: delegation_id, tar_id, authority_budget

Phase 5 – Verification
Any party runs: python omnix_atf_verify.py receipt.json
DR signature verified against embedded public key
TAR status confirmed ADMITTED
execution_ref cross-checked with GovernanceReceipt.receipt_id
Chain root traced to Tier-1 human principal
Result: VERIFIED or NOT VERIFIED

```

## 2.3 Authority Budget Arithmetic

The authority budget is a real number in  $[0.0, 100.0]$  representing the fraction of full authority held by an agent. The Monotonic Authority Reduction (MAR) invariant requires:

```

For all DRs D in the trust lattice:
D.authority_budget_granted ≤ D.authority_budget_delegator

For all chains C = [DR1, DR2, ..., DRn]:
DR1.authority_budget_granted ≥ DR2.authority_budget_granted ≥ ... ≥
DRn.authority_budget_granted

```

This invariant is formally specified in TLA+ and verified by model checking over bounded state spaces (ATF-FV-1.0). It ensures that delegation cannot *create* authority — only distribute and bound it.

### Example delegation chain:

```

HUMAN-TIER1 budget = 100.0 (full authority)
■■ ATFDR-ROOT → AID-FINANCE-001 budget = 80.0 (20% reduction)
■■ ATFDR-A → AID-FINANCE-002 budget = 50.0 (37.5% reduction)
■■ ATFDR-B → AID-FINANCE-003 budget = 20.0 (60% reduction)

```

### 3. Formal Invariants

ATF-1.0 defines six formally specified invariants. All are mandatory for ATF-COMPLIANT-LEVEL-2 and above.

| Invariant                     | Identifier  | Formal Statement                                                      |
|-------------------------------|-------------|-----------------------------------------------------------------------|
| Monotonic Authority Reduction | ATF-INV-001 | $DR.budget\_granted \leq DR.budget\_delegator$ for all DRs            |
| Acyclicity                    | ATF-INV-002 | The trust lattice is a DAG — no delegation cycle exists               |
| Chain Root Consistency        | ATF-INV-003 | All DRs in a chain share the same chain_root_id                       |
| Content Hash Immutability     | ATF-INV-004 | DR fields are immutable post-issuance (hash binds all fields)         |
| Temporal Non-Future-Dating    | ATF-INV-005 | $TAR.execution\_ns \leq \text{current time at verification}$          |
| Independent Verifiability     | ATF-INV-006 | Full chain verifiable offline using only receipts and root public key |

**TLA+ Coverage:** INV-001 (MARInvariant), INV-002 (AcyclicityInvariant), INV-003 (ChainRootConsistency), and INV-004 (ImmutabilityProperty) are formally specified and model-checked in docs/formal/ATF-TLA-SPEC.tla. This is the same formal verification methodology used by Amazon Web Services for DynamoDB.

### 4. Cryptographic Specification

#### 4.1 Signature Algorithm

| Property        | Value                                           |
|-----------------|-------------------------------------------------|
| Algorithm       | ML-DSA-65 (Dilithium-3)                         |
| Standard        | NIST FIPS 204 (August 2024)                     |
| Security level  | NIST PQC Level 3                                |
| Public key size | 1,952 bytes                                     |
| Signature size  | 3,293 bytes                                     |
| Hash function   | SHA-256 (content hash construction)             |
| Encoding        | Base64 standard (signature), hex (content hash) |

#### 4.2 Content Hash Construction

All ATF artifacts use a deterministic content hash for signature binding:

```
canonical = json.dumps(fields, sort_keys=True, separators=(',', ':'), ensure_ascii=True)
content_hash = hashlib.sha256(canonical.encode('utf-8')).hexdigest()
```

| Artifact | Hashed Fields                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DR       | delegation_id, delegator_id, delegate_id, task_scope, authority_budget_granted, authority_budget_delegator, chain_root_id, delegation_depth, expires_at, status, created_at   |
| TAR      | tar_id, delegation_id, agent_id, execution_ref, execution_ns, dr_status_at_admission, dr_expires_at, authority_budget, domain, task_action, admission_status, chain_root_id   |
| DTR      | dtr_id, source_delegation_id, source_domain, target_domain, source_agent_id, target_agent_id, source_authority_budget, translated_budget, translation_discount, chain_root_id |

### 4.3 Fallback Mode

When ML-DSA-65 is unavailable, ATF falls back to content hash only (no PQC signature). This mode (`pqc_signature = null`) is explicitly: **Permitted** for ATF-COMPLIANT-LEVEL-1 development/test environments; **Prohibited** for ATF-COMPLIANT-LEVEL-2+ production deployments.



## 5. Temporal Admissibility (ADR-157)

The Temporal Admissibility Record (TAR) is ATF's mechanism for proving that an agent's authority was valid at the exact moment of execution.

### 5.1 Why TAR Is Necessary

JWT exp claims verify that a token is not expired at the time of checking. This is an implicit check — there is no artifact proving the check occurred. TAR makes the check explicit and produces a signed, persistent record:

| Property                  | JWT exp check | TAR (ATF-INV-005)                 |
|---------------------------|---------------|-----------------------------------|
| Existence check           | Yes           | Yes                               |
| Signed artifact           | No            | Yes (ML-DSA-65)                   |
| Timestamp granularity     | Second        | Nanosecond-resolution             |
| Persistence               | No            | Yes (database row, immutable)     |
| Binding to execution      | No            | Yes (execution_ref)               |
| Independent verifiability | No            | Yes (offline CLI)                 |
| Rejection record          | No            | Yes (REJECTED status with reason) |

### 5.2 TAR Issuance Protocol

```
Input: DR, agent_id, task_action, execution_ref
Output: TemporalAdmissibilityRecord (ATFTAR-{16HEX})

1. execution_ns ← time.time_ns() # Nanosecond capture – BEFORE checks
2. Verify DR.status == ACTIVE
3. Verify execution_ts < DR.expires_at
4. Verify execution_ts >= DR.created_at
5. Set admission_status ← ADMITTED (or REJECTED with reason)
6. Compute content_hash over all TAR fields including execution_ns
7. Sign content_hash with ML-DSA-65
8. Persist TAR to atf_temporal_records table
9. Return TAR
```

**Step 1 is executed before steps 2–4.** This ensures the execution\_ns reflects the actual admission time, not the outcome of the checks. The TAR documents the authority state *at admission*, before any decision logic executes.

## 6. Cross-Domain Trust Portability (ADR-158)

In multi-domain deployments, an agent authorized in domain FINANCE may need to access resources in domain HEALTHCARE. Cross-domain trust requires explicit authority translation with mandatory reduction.

6.1 Domain Translation Receipt (DTR / ATFDTR-{{16HEX}})

```
source_budget = 60.0 (FINANCE domain)
discount = 0.30 (FINANCE → HEALTHCARE policy)
translated_budget = 60.0 × (1 - 0.30) = 42.0 (HEALTHCARE domain)
```

The discount is mandatory — cross-domain translation **MUST** reduce authority (CDTP-INV-003). An agent cannot gain authority by crossing domains.

6.2 Standard Domain-Pair Discount Policies

| Source → Target      | Default Discount | Rationale                        |
|----------------------|------------------|----------------------------------|
| HEALTHCARE → FINANCE | 30%              | High sensitivity of both domains |
| FINANCE → HEALTHCARE | 30%              | Symmetric high-sensitivity       |
| FINANCE → INSURANCE  | 15%              | Adjacent regulatory domain       |
| DEFENSE → FINANCE    | 40%              | Strict isolation requirement     |
| DEFENSE → HEALTHCARE | 45%              | Maximum cross-domain sensitivity |
| Other pairs          | 20%              | Default cross-domain discount    |

## 7. Governance Receipt Integration

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### 7.1 The Triple Chain

Every OMNIX governance decision originating from an ATF-registered agent produces three cross-referenced artifacts:

```
{
  "receipt_id": "OMNIX-FIN-20260512-A3F7B2",
  "decision": "APPROVED",
  "pqc_signature": "ML-DSA-65 signature ...",

  "atf_context": {
    "delegation_id": "ATFDR-8B2C4D6E1F3A5B7C",
    "tar_id": "ATFTAR-C4D8E2F1A3B5C7D9",
    "agent_id": "AID-FINANCE-3A7F9B2C1D4E5F6A",
    "delegator_id": "HUMAN-TIER1-HN-001",
    "admission_status": "ADMITTED",
    "execution_ns": 1747058400000000000,
    "authority_budget": 60.0,
    "chain_root_id": "ATFDR-8B2C4D6E1F3A5B7C",
    "pqc_signed": true
  }
}
```

This structure enables a verifier to answer all four questions using only this JSON document and the root public key.

### 7.2 Verification Flow

```
verify(receipt.json):
1. Check receipt.pqc_signature over receipt.content_hash → receipt integrity
2. Fetch DR by atf_context.delegation_id → authority claim
3. Verify DR.pqc_signature over DR.content_hash → delegation integrity
4. Verify DR.delegator_public_key signed the hash → principal identity
5. Fetch TAR by atf_context.tar_id → temporal proof
6. Verify TAR.pqc_signature over TAR.content_hash → TAR integrity
7. Check TAR.admission_status == ADMITTED → execution admitted
8. Check TAR.execution_ref == receipt.receipt_id → binding holds
9. Traverse chain to chain_root_id → human traceability
Result: VERIFIED | NOT VERIFIED

All steps can be performed offline using only the receipt artifacts.
```

## 8. Independent Verifiability (ATF-INV-006)

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*"Any party MUST be able to verify a delegation chain using only the receipts and the root public key. No access to the issuing platform, API, account, or internet connection is required."*

This property distinguishes ATF from platform-dependent audit trails. A regulator conducting a post-incident review can:

- Download the receipt artifacts from the institution's disclosure
- Run `python omnix_atf_verify.py receipt.json`
- Receive a complete VERIFIED/NOT VERIFIED verdict with full chain analysis
- Without any interaction with OMNIX or the deploying institution's systems

This is analogous to how anyone can verify a PGP-signed document without contacting the key issuer — but applied to the entire AI agent authority chain.

## 9. Compliance Framework

ATF defines three compliance levels:

### Level 1 — Basic (Development and Proof-of-Concept)

- DR content hash (SHA-256)
- MAR enforcement
- Chain traversal
- PQC signature: SHOULD (not required)
- TAR: OPTIONAL

### Level 2 — Standard (Production, Enterprise)

- All Level-1 requirements
- PQC signature REQUIRED (ML-DSA-65)
- TAR REQUIRED for all governance-connected executions
- ATF CCS  $\geq 80$  required for high-assurance domains
- Independent verifiability demonstrated

### Level 3 — Sovereign (Regulated, Critical Infrastructure)

- All Level-2 requirements
- RFC 3161 TSA timestamp counter-signature on TARs
- FIPS 140-3 validated cryptographic module SHOULD be used
- Formal verification documentation (TLA+ or equivalent)
- Public CLI verifier available for independent audit
- Cascade revocation within 1 hour of compromise detection

## 10. Interoperability

| Framework / Protocol       | Relationship to ATF                                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| W3C Verifiable Credentials | ATF DRs share structural concepts. ATF extends with authority budget arithmetic, MAR invariant, TAR as separate signed artifact, and TLA+-verified formal invariants. |
| IETF JWT (RFC 7519)        | ATF receipts can be serialized as JWT claims. The exp claim $\approx$ DR.expires_at. TAR provides stronger evidence than exp — signed artifact, not just a claim.     |
| OpenID Connect             | ATF agent identities can be registered as OAuth 2.0 clients. chain_root_id can be used as sub claim linking to human principal's OIDC identity.                       |
| SPIFFE/SPIRE               | ATF extends workload identity with delegation chain, authority arithmetic, and governance receipt integration.                                                        |

| Framework / Protocol         | Relationship to ATF                                                                                                        |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| LangChain / AutoGen / CrewAI | ATF is framework-agnostic. ATFConnector integrates into any Python-based agent framework without modifying existing logic. |

## 11. Implementation Reference

The OMNIX ATF reference implementation consists of the following functional components. Internal module structure is proprietary; public interfaces are exposed via the REST API and the offline CLI verifier.

| Component                 | Role                                                                                                     | Public Interface                           |
|---------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Trust Lattice Engine      | DAG management, AgentIdentity registration, DelegationReceipt issuance, MAR enforcement                  | /api/atf/register · /api/atf/delegate      |
| Temporal Authority Engine | TAR issuance and verification, nanosecond-resolution admission capture, ADMITTED/REJECTED status         | /api/atf/admit · /api/atf/temporal/verify  |
| Cross-Domain Bridge       | DTR issuance, domain-pair discount policy enforcement, CDTF invariant verification                       | /api/atf/translate                         |
| Governance Connector      | Integration layer binding ATF context to GovernanceReceipt — non-blocking, backward compatible           | Embedded in governance pipeline (internal) |
| REST API Layer            | Full ATF lifecycle endpoints — register, delegate, admit, verify, chain traversal                        | /api/atf/* (documented in RFC-ATF-1)       |
| Offline CLI Verifier      | Standalone verifier — DR, TAR, chain, receipt modes. Zero platform dependency (ATF-INV-006)              | omnix_atf_verify.py (public, open source)  |
| Formal Specification      | TLA+ model — 5 invariants, bounded state space verification, MAR + Acyclicity + ChainRoot + Immutability | ATF-TLA-SPEC.tla (public repository)       |

## 12. Known Limitations and Open Questions

| Limitation                              | Mitigation                                             | Status  |
|-----------------------------------------|--------------------------------------------------------|---------|
| Clock trust in TAR (TM-004)             | RFC 3161 TSA integration (planned Level-3 requirement) | OPEN    |
| Revocation propagation latency (TM-006) | Short DR validity + centralized revocation registry    | PARTIAL |
| Theorem-proved MAR vs model-checked     | Port to Coq/Lean for one invariant (planned)           | PLANNED |
| FIPS 140-3 validated library            | Library-agnostic spec; swap pqc for validated lib      | OPEN    |
| Multi-instance TAR uniqueness           | Redis SETNX / DB unique constraint                     | PARTIAL |

## 13. Related Work

**W3C Verifiable Credentials:** Identity claims with PQC extensions (emerging drafts). No authority budgets or temporal admissibility artifacts.

**SPIFFE/SPIRE:** Workload identity for service meshes. No delegation chains or authority arithmetic.

**OpenID Connect:** Authentication protocol. No authorization governance chain.

**NIST SP 800-207 (Zero Trust):** Zero Trust Architecture principles. ATF implements explicit trust verification before each execution event.

**NIST AI RMF (January 2023):** AI Risk Management Framework. ATF directly implements accountability requirements in the GOVERN function.

**EU AI Act (August 2024):** ATF addresses Arts. 13/14 (transparency and human oversight) for high-risk AI systems.

**CRYSTALS-Dilithium / ML-DSA:** NIST FIPS 204 (August 2024). ATF adopts ML-DSA-65 as mandatory signature algorithm for Level-2+ compliance.

## 14. Conclusion

The OMNIX Agent Trust Fabric addresses a specific, well-defined gap in AI governance infrastructure: the absence of cryptographic proof of *who authorized an AI agent, under what authority bound, when that authority was valid, and whether execution was admitted at that exact moment*.

ATF provides this through a three-artifact chain (DR + TAR + GovernanceReceipt), formally specified invariants (TLA+ model-checked), post-quantum cryptography (ML-DSA-65, NIST FIPS 204), and independent offline verifiability (ATF-INV-006).

The design is complementary to existing orchestration frameworks, not a replacement for them. It operates at the governance layer that precedes execution — making the authority chain verifiable before, during, and after an AI agent acts.

As autonomous AI systems take on greater responsibility in enterprise and regulated environments, this governance layer becomes a prerequisite for institutional trust. OMNIX ATF provides the formal specification, reference implementation, and public verification tooling to establish that trust on a cryptographic foundation.



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|---------------------------|-----------------------|------------------|------------------------------------------------------------|
| <b>Document ID:</b> OMNIX |                       | <b>Date:</b> May | <b>Classification:</b> Public — Institutional Distribution |
| -WP-ATF-2026-001          | <b>Version:</b> 1.0.0 | 2026             |                                                            |